

Route or Distill? Investigating Lightweight Perceptual Skill Routing for Visual Quadruped Locomotion

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Abstract—Integrating visual locomotion skills enables quadrupeds to traverse diverse obstacles, but distilling specialist behaviors into a unified policy requires additional simulated rollouts and policy optimization. We ask how simply an existing specialist library can be integrated while retaining locomotion capability. We investigate lightweight terrain-based routing that composes frozen specialists through a supervised classifier trained on terrain-labeled trajectories, requiring no additional locomotion-policy optimization. Using the same specialist library, we compare routing with a Dagger-style distilled controller and ground-truth routing. Within the routing framework, we examine geometric depth features and a raw-depth convolutional network paired with instantaneous selection, exponential moving averaging, or Bayesian filtering. Experiments connect offline recognition, sequential selection, and closed-loop simulated locomotion, revealing that stronger classification accuracy and switching stability do not necessarily predict better locomotion performance. On held-out mixed-terrain courses, the best evaluated router achieves 92.6% episodic success, compared with 74.3% for the evaluated distilled checkpoint and 99.1% for ground-truth routing. These results support simple post-hoc routing as a practical integration alternative and emphasize the importance of evaluating perception and temporal selection through their closed-loop consequences. Anonymized Project Website https://o28659645-png.github.io/route_distill_icra_2027/.

I. INTRODUCTION

Quadruped robots have demonstrated remarkable advances in mobility driven by their potential to operate in unstructured environments [1]–[6]. Quadruped robots operating in such environments must negotiate qualitatively distinct obstacles. Jumping over gaps in the terrain require controlled takeoff and landing, raised obstacles require climbing and descent, and stairs require repeated contact transitions (Fig. 2). A useful robot must respond to the environment along a route by combining these capabilities during deployment using onboard perception. Therefore, learning specialized visual locomotion skills is only part of the problem; the system must also determine which skill to execute and when. Once specialist policies have acquired these capabilities, the remaining problem is integration; should their behaviors be consolidated into a new policy, or can they be retained and selected through a simple routing layer?

Recent works have explored an increasing popular training paradigm: multi-skill policy distillation [5]–[9]; which consolidates specialist behavior into a single, unified policy through additional online, supervised training. This produces a single deployable policy but requires an additional round of expensive simulated training and optimization. At the same time, matching specialist outputs during training does not guarantee preservation of their closed-loop performance.

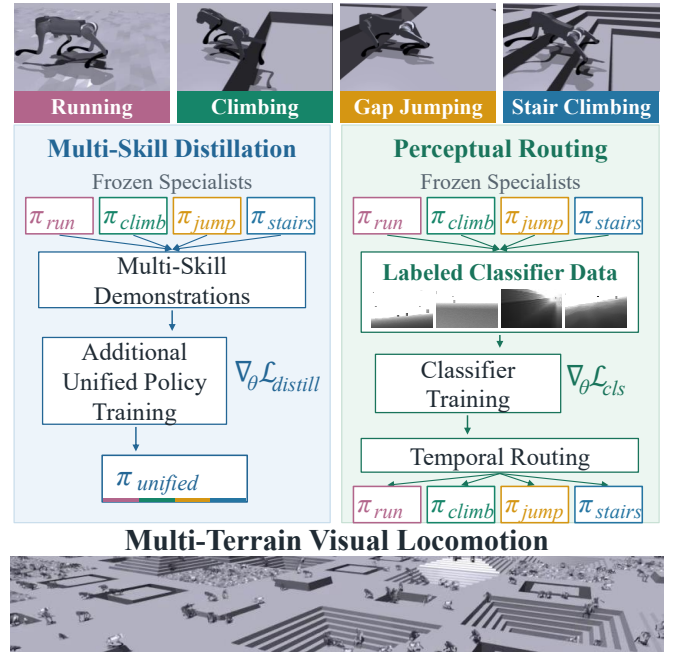


Fig. 1. Overview of the two multi-skill integration strategies studied in this work. Distillation uses frozen specialist policies as teachers to generate multi-skill demonstrations and trains a new unified locomotion policy through additional student optimization. Routing preserves the specialist policies and instead trains a lightweight perception model whose predictions are temporally filtered and used to select the active skill online. We compare these alternatives in terms of closed-loop locomotion performance, routing behavior, additional training cost after specialist acquisition, and deployment overhead.

Moreover, an existing specialist library also enables a simpler integration strategy: learn which policy to execute while retaining the specialists themselves. Skill selection/routing avoids further locomotion-policy optimization, although it still requires training a selector and managing the transitions between skills. In this work, we ask whether lightweight post-hoc routing provides a simple and practical alternative to distillation for integrating an existing library of visual locomotion skills? We examine this question through both locomotion performance and integration effort: wherein routing retains the specialist library and adds a selector and distillation consolidates behavior into one policy. Consequently, potential savings in data collection and training must be considered alongside inference and memory costs at deployment.

We investigate lightweight terrain-based routing as a simple alternative to distilling an existing library of locomotion specialists into a unified policy. Our routed system composes frozen specialists through a supervised classifier trained on terrain-labeled simulation trajectories, requiring no additional

locomotion-policy optimization. Through matched comparisons of geometric depth features and a raw-depth CNN, each paired with instantaneous selection, exponential moving averaging, or Bayesian filtering, we examine how perception and temporal processing affect routing. These choices intentionally keep the integration pipeline simple. Although more sophisticated learning frameworks could support richer skill-selection behavior, our goal is to determine whether simple classifiers and lightweight post-hoc temporal processing suffice to match or exceed the locomotion performance of the evaluated distilled controller, while measuring the integration and inference costs of both approaches.

In our experiments, we connect instantaneous classification, sequential selection, and closed-loop locomotion, showing that stronger offline recognition and switching metrics do not necessarily yield better locomotion performance. Same-trajectory replay and transition-aligned analysis investigate this discrepancy through the stability, anticipation, and responsiveness of skill selection. Finally, we compare routed specialists with ground-truth routing and a distilled controller built from the same specialist library across held-out terrain sequences and difficulty levels. By jointly evaluating locomotion performance, routing behavior, deployment inference cost, and additional data-collection and training costs, we characterize when simple post-hoc routing provides a practical alternative to multi-skill distillation. Our findings suggest that such systems serve as a viable alternative to unified policy distillation for complex visual locomotion tasks and deserves further investigation.

II. RELATED WORK

Visual Locomotion and Integration Through Distillation Policy distillation [7] consolidates specialist behaviors into a unified controller, while DAgger [10] provides expert supervision on learner-visited states. This paradigm has been applied to quadruped locomotion through obstacle-specific specialists [5], sometimes followed by RL fine-tuning for improved generalization [6]. Recent humanoid systems similarly distill visual specialists using DAgger and RL [8] or adaptive specialist selection [9]. In contrast, we study lightweight terrain-based routing of a fixed skill library, comparing closed-loop performance and post-specialist integration cost. Unified visual parkour policies trained directly across diverse obstacles [3], [11] are therefore outside our experimental comparison.

Perception-Driven Skill Selection Visual perception has long been used to select among reusable robot behaviors. Visual meta-policies can sequence low-level skills across platforms [12], while quadruped systems have learned terrain-dependent speed and gait selection [13] or used reconstructed geometry to select and command specialized locomotion skills [4]. VOCALoco [14] similarly composes pretrained specialists by predicting policy viability and energetic cost from local terrain. We instead study whether simple terrain classification with lightweight temporal processing is sufficient for integrating a fixed specialist library, with emphasis on matched routing–distillation comparisons and

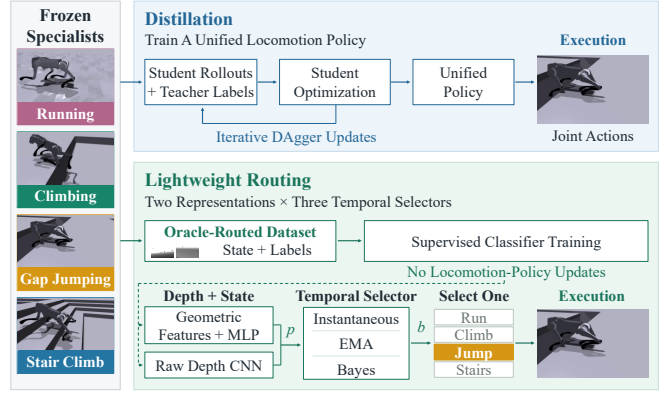


Fig. 2. Comparison of multi-skill integration strategies. Distillation uses frozen specialists to train a unified policy through additional policy optimization, while our routing approach preserves the specialist library and selects skills online using terrain classification with temporal smoothing.

on how perception and switching behavior affect closed-loop locomotion and integration cost.

Temporal Inference for Skill Routing. Temporal structure has long been used to stabilize robot skill selection under uncertain observations. Prior work combines classifier outputs with learned sequence models [15], hidden Markov or semi-Markov representations [16], [17], and Bayesian filtering for terrain-dependent gait adaptation [18]–[20]. More recent legged-robot approaches use recurrent or attention-based temporal models to improve terrain recognition and robustness [21], [22]. We compare instantaneous selection, EMA smoothing, and Bayesian filtering to study how temporal processing affects switching stability, anticipation, and transition responsiveness when routing among frozen locomotion specialists, and whether these simple mechanisms are sufficient for a practical alternative to multi-skill distillation.

III. APPROACH

A. Problem Formulation

Visual Locomotion Skill Learning At each control step t , a quadruped robot receives an proprioceptive observation $\mathbf{o}_t$ comprising joint positions and velocities, body angular velocity, projected gravity, velocity commands, and the previous action. Visual information arrives through an asynchronous, lower-rate depth stream: the most recently available image $\mathbf{D}_{\kappa(t)}$, subject to sensing latency and held between camera updates, is encoded by a CNN into a visual representation $\mathbf{d}_t$. A DreamWaQ-style history encoder [2] maps a short history of the H latest observations to a latent context $\mathbf{z}_t$ and an estimate $\hat{\mathbf{v}}_t$ of body-frame linear velocity. The policy $\pi_{\theta}(\mathbf{a}_t | \mathbf{o}_t, \mathbf{z}_t, \hat{\mathbf{v}}_t, \mathbf{d}_t)$ produces joint-position actions whose scaled offsets from the nominal stance define targets for PD control.

We formulate this visual locomotion problem as a partially observable Markov decision process with underlying state $\mathbf{s}_t$, transition dynamics $T(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)$, and task reward r , and optimize the expected discounted return, $\max_{\theta} \mathbb{E}_{\pi_{\theta}} [\sum_{t=0}^{\infty} \gamma^t r(\mathbf{s}_t, \mathbf{a}_t)]$, where $\gamma \in [0, 1)$. Training uses proximal policy optimization [23] with an asymmetric critic

that additionally accesses privileged simulator information, including true base velocity, contact states, local terrain heights, and randomized dynamics parameters. Following Zhuang et. al. [5], we train a rough-terrain base policy and fine-tune separate specialists for gap crossing, obstacle climbing, and stair traversal. The resulting policies form a shared library that we subsequently integrate through terrain-based routing or distillation into a unified controller.

Multi-Skill Policy Distillation. Following the DAgger distillation procedure in [5], we consolidate the frozen specialist library into a single visual policy. The unified policy executes its deterministic actions during simulated rollouts, while the simulator’s ground-truth terrain mapping identifies the specialist that supplies an action target at each visited state. Let $\mathbf{x}$ be the current observation, observation history, and latest available depth image, and let $k^*(\mathbf{x})$ denote the teacher assigned to that sample. We alternate student rollout collection with supervised updates that minimize

$$\mathcal{L}_{\text{distill}}(\theta) = \mathbb{E}_{\mathbf{x} \sim d_{\pi_\theta}} [\|\pi_\theta(\mathbf{x}) - \pi_{k^*(\mathbf{x})}(\mathbf{x})\|_1], \quad (1)$$

where d_{π_θ} denotes the distribution of inputs encountered under student control and $\pi(x)$ denotes a policy’s deterministic action output. The ℓ_1 norm sums absolute errors across action dimensions, and the expectation is approximated using teacher-labeled rollout batches. Student and teacher evaluate the same input histories and depth observations through their respective encoders, allowing supervision on states induced by the student’s own behavior. The student’s actor, history encoder, and visual CNN are optimized jointly through action imitation, with all specialists frozen and no reward-based policy optimization during distillation.

The resulting controller executes the integrated behaviors through a single visual policy without an explicit skill selector. Distillation requires repeated simulated rollouts and supervised policy updates, adding substantial computation and training time after specialist acquisition. We therefore ask whether comparable locomotion performance can be achieved with a simpler and less costly integration procedure: retaining the frozen specialists and learning when to execute each one through simple terrain perception and lightweight temporal filtering.

B. Simple Depth Perception for Terrain Classification

To simplify post-specialist integration, we cast skill selection as supervised terrain recognition while keeping all locomotion policies frozen. Ground-truth-routed rollouts on mixed-terrain courses provide normalized 48×64 depth images, body orientation, angular velocity, and terrain labels for four classes: rough terrain, gaps, stairs, and climbing obstacles. Using the same roll, pitch, and angular-velocity inputs, we compare an MLP over compact geometric depth features with a CNN operating directly on depth images. The engineered descriptors summarize discontinuities, roughness, and spatial layout, allowing us to test whether explicit geometric structure and limited labeled rollouts are sufficient for routing, or whether learned depth representations provide additional control-relevant information.

Geometric Depth Features. For the preprocessed depth signal $\mathbf{D}$, we first perform Gaussian-smoothing resulting in $\mathbf{I} = \mathbf{G} * \mathbf{D}$. We then summarize Sobel gradients $\nabla_S(\mathbf{I})$ [24] through their magnitudes, directions, signed transitions, and row-wise repetition, providing cues for obstacle boundaries and repeated stair structures. Laplacian magnitudes $|\Delta(\mathbf{I})|$ and multiscale high-pass statistics $|\mathbf{I} - \mathbf{A}_s * \mathbf{I}|$ capture complementary depth variation, following established multiscale image representations [25]; where $\mathbf{A}_s$ denotes local averaging at scale s . Depth dispersion, near–far differences, validity boundaries, and regional fractions of close, distant, or invalid returns provide additional obstacle and gap cues. Selected statistics are also pooled over a 2×3 grid to retain coarse spatial layout. Concatenating these descriptors with roll, pitch, and body angular velocity yields a 54-dimensional vector, standardized using training-set statistics. We train an MLP with three hidden layers of 512, 256, and 128 units, ELU activations, and a four-class linear output layer.

Learned Depth Representation. A CNN learns terrain-discriminative visual features directly from depth images, allowing spatial cues omitted by geometric summaries to inform skill selection. We adapt the convolutional depth-encoding design of [5] into a compact, feedforward terrain classifier. Normalized 48×64 depth images $\mathbf{D}$ pass through two convolutional layers with 8 and 16 channels, 5×5 and 3×3 kernels, ELU activations, and 2×2 max pooling after the first convolution. The flattened visual features are concatenated with roll, pitch, and body angular velocity, then processed by a 128-unit hidden layer and a four-class linear output layer. The CNN is intentionally shallow and compact to limit computation and memory requirements on robot hardware while allowing visual features to be learned directly from depth.

C. Lightweight Temporal Processing for Visual Skill Routing

Instantaneous terrain classification ignores observation history, making skill selection susceptible to fluctuations caused by sensor noise, viewpoint changes, and ambiguous terrain boundaries. Temporal processing can suppress these fluctuations, but excessive persistence may delay transitions needed for successful locomotion. We investigate lightweight post-hoc filters that operate on classifier outputs while keeping the classifiers and locomotion specialists frozen, incorporating history without training an additional sequence model. This preserves the simplicity of our integration pipeline and tests the lower-limits on how much temporal reasoning effective routing requires. We first consider exponential moving averaging, which smooths successive predictions, and then Bayesian filtering, which combines current evidence with a simple fixed-probability terrain-transition model to recursively update class beliefs.

Exponential Moving Averaging. As a lightweight temporal baseline, we smooth classifier logits ℓ_t using

$$\bar{\ell}_t = \alpha \ell_t + (1 - \alpha) \bar{\ell}_{t-1}, \quad \alpha \in (0, 1], \quad (2)$$

initialized with the first prediction. The selected skill corresponds to $\arg \max_k \bar{\ell}_{t,k}$. EMA suppresses short-lived

fluctuations with minimal computation and memory, but does not explicitly model terrain-transition probabilities. Residual fluctuations can therefore still produce false switches, particularly when competing classes have similar scores; reducing α strengthens smoothing but can delay genuine transitions. This stability–responsiveness tradeoff motivates Bayesian filtering, which combines new classifier evidence with a predicted terrain-state belief under an explicit transition model.

Bayesian filtering. We model terrain continuity through a first-order Markov process and recursively update a belief $\mathbf{b}_t$ over the K terrain classes $b_t(k)$. The fixed transition matrix $\mathbf{T}$ assigns $T_{kk} = s$ to remaining in the same class and $T_{jk} = (1-s)/(K-1)$ to each alternative, where $T_{jk} = P(k_t = k \mid k_{t-1} = j)$. We use $K = 4$ and $s = 0.9$, encoding persistence without assuming a particular terrain ordering. Starting from a uniform belief, each classification step performs prediction and correction:

$$\begin{aligned}\tilde{b}_t(k) &= \sum_j T_{jk} b_{t-1}(j), \\ b_t(k) &= \frac{p_t(k) \tilde{b}_t(k)}{\sum_i p_t(i) \tilde{b}_t(i)},\end{aligned}\tag{3}$$

where classifier probabilities $p_t(k)$ serve as observation-likelihood surrogates. The selected skill is $\arg \max_k b_t(k)$. Compared with fixed-rate logit averaging, this prediction–correction structure explicitly models persistence while allowing new evidence to reinforce or overturn the predicted belief. It can suppress isolated contradictory predictions, while transition probabilities retain support for alternative classes. These properties offer a different stability–responsiveness tradeoff from EMA, although stronger persistence can also delay genuine switches.

D. Training and Deployment Details

Training details. We train locomotion specialists in Isaac Gym [26] using PPO, first learning a rough-terrain baseline and then fine-tuning for gaps, stairs, and climbing obstacles. The classifier dataset contains [N_classifier] labeled observations, while temporal evaluation uses [N_sequential] chronologically ordered observations. Datasets are partitioned into training, validation, and test sets using a 60/20/20 split over environment groups. Both classifiers use identical partitions, Adam optimization, and validation-loss early stopping across three training seeds. Classifier and temporal-filter hyperparameters are selected through structured validation-set grid searches and held fixed during final evaluation. Distillation alternates student-controlled rollouts with supervised updates minimizing the summed absolute action error against frozen, terrain-selected specialists. The student’s actor, history encoder, and visual encoder are optimized jointly through pure imitation.

Inference During simulated locomotion, policy inference runs at 50 Hz using current proprioception, observation history, and the latest asynchronously available depth image. Routing updates run at 10 Hz, with the selected specialist retained between updates, as seen in Alg. 1. The distilled

Algorithm 1: Deployment of frozen specialists with terrain routing

Input: Specialists $\{\pi_k\}$, perception pipeline C , temporal selector F

- 1 Initialize observation history and temporal state;
- 2 Select the rough-terrain skill k ;
- 3 **while** *controller is active* **do**
- 4 Read proprioception and latest depth image D ;
 update history;
- 5 $\mathbf{a} \leftarrow \pi_k(\mathbf{o}, \mathbf{o}^H, D)$;
- 6 Apply action scaling and command limits; publish PD targets;
- 7 **if** *routing update is due* **then**
- 8 $\ell \leftarrow C(D, \text{roll}, \text{pitch}, \omega)$;
- 9 **if** F *is instantaneous* **then**
- 10 $\mathbf{v} \leftarrow \ell$;
- 11 **else if** F *is EMA* **then**
- 12 $\mathbf{v} \leftarrow \text{EMAUPDATE}(\ell)$;
- 13 **else**
- 14 $\mathbf{v} \leftarrow \text{BAYESUPDATE}(\text{softmax}(\ell))$;
- 15 **end**
- 16 $k \leftarrow \arg \max_j v_j$;
 // Use the selected skill at the next policy step
- 17 **end**
- 18 **end**

controller uses the same observation and action interfaces, replacing specialist selection with a single student policy. See the appendix for full training and implementation details, including per-skill PPO configurations, terrain ranges, dataset construction, hyperparameter-search spaces and selection criteria, and deployment settings.

IV. EVALUATION

We evaluate post-hoc skill routing through three complementary experiments: offline terrain classification, offline sequential selection, and closed-loop locomotion on held-out mixed-terrain courses. These experiments address the following questions:

- Q1. What tradeoffs arise between perceptual complexity, terrain-recognition accuracy, and classifier training cost?
- Q2. How does temporal processing affect the stability and responsiveness of skill selection as terrain changes?
- Q3. How does specialist routing compare with ground-truth routing and distillation in locomotion performance and integration cost, and can offline routing metrics fail to predict closed-loop performance?

A. Instantaneous Classification

We first examine how perception representation affects terrain recognition at a fixed data budget. The geometric-feature MLP and raw-depth CNN use the same environment-disjoint dataset partitions, depth observations, and robot-state

TABLE I

INSTANTANEOUS TERRAIN CLASSIFICATION. HIGHER ACCURACY AND F1 SCORES AND LOWER PROBABILISTIC LOSSES INDICATE BETTER PERFORMANCE. BEST VALUES ARE BOLD.

Representation	Balanced acc. (%) $\uparrow$	Macro-F1 (%) $\uparrow$	NLL $\downarrow$
Geometric Features	95.04 ± 0.18	95.15 ± 0.19	0.129 ± 0.005
Raw Depth	92.81 ± 0.21	93.17 ± 0.14	0.178 ± 0.004

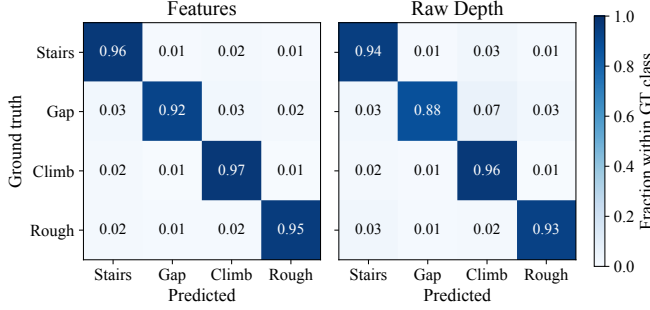


Fig. 3. Row-normalized confusion matrices for geometric-feature and raw-depth classifiers.

inputs across four terrain classes. Each frame is classified independently, excluding temporal filtering. This comparison examines whether explicit geometric structure sufficiently simplifies learning for effective routing, or whether learned image features capture control-relevant spatial information omitted by the engineered summaries. Recognition results are reported as mean $\pm$ standard deviation across three classifier training seeds.

Table I reports balanced accuracy (mean class recall), macro-F1 (mean class F1), and negative log-likelihood (NLL), which penalizes low probability assigned to the true class. Geometric features improve balanced accuracy by 2.23 percentage points and macro-F1 by 1.98 percentage points, while reducing NLL from 0.178 to 0.129. The confusion matrices in Fig. 3 show higher recall across all four classes. The largest improvement occurs for gaps, with recall increasing from approximately 88% to 92%; gap observations misclassified as climbing terrain decrease from approximately 7% to 3%. Gaps nevertheless remain the least accurately recognized class for both representations. These results are consistent with the structural inductive bias supplied by geometric descriptors. Explicit representations of discontinuities, roughness, and spatial layout expose cues such as repeated stair edges and gap boundaries, reducing the geometric structure that must be learned from raw images.

B. Sequential Terrain Prediction

We next examine how temporal processing trades selection stability against responsiveness. Both perception representations are evaluated with instantaneous selection, EMA, and Bayesian filtering on the same 720,000 held-out ordered frames, containing 6,829 ground-truth terrain transitions. Identical observations isolate selector behavior from differences in robot trajectories. All methods share

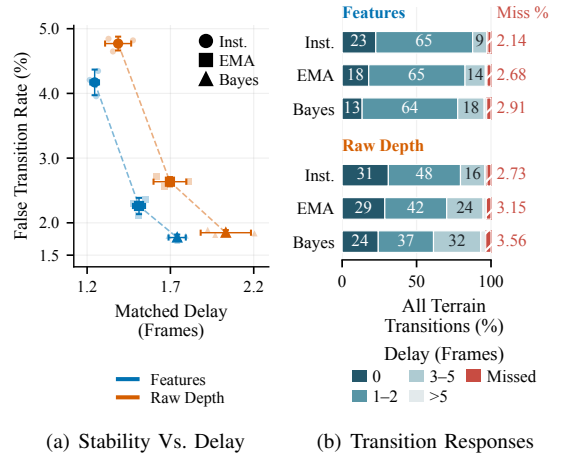


Fig. 4. Sequential prediction stability and transition responsiveness across three classifier seeds.

TABLE II

SEQUENTIAL TERRAIN PREDICTION (MEAN $\pm$ STANDARD DEVIATION ACROSS THREE TRAINING SEEDS). BEST VALUES ARE BOLD.

Representation	Selector	Balanced acc. (%) $\uparrow$	False transitions (%) $\downarrow$	Delay (frames) $\downarrow$	Miss rate (%) $\downarrow$
Features	Instantaneous	95.04 ± 0.18	4.17 ± 0.20	1.25 ± 0.03	2.14 ± 0.06
	EMA	95.99 ± 0.14	2.26 ± 0.13	1.51 ± 0.04	2.68 ± 0.11
	Bayes	96.12 ± 0.08	1.77 ± 0.05	1.74 ± 0.05	2.91 ± 0.06
Raw depth	Instantaneous	92.81 ± 0.21	4.77 ± 0.11	1.39 ± 0.08	2.73 ± 0.05
	EMA	93.94 ± 0.25	2.64 ± 0.07	1.70 ± 0.10	3.15 ± 0.03
	Bayes	94.34 ± 0.21	1.85 ± 0.04	2.03 ± 0.15	3.56 ± 0.08

sequence boundaries and reset rules, with filter state cleared between sequences. Results report mean $\pm$ standard deviation across three classifier training seeds.

Alongside balanced accuracy, we measure false-transition rate: predicted label changes occurring while the ground-truth label remains unchanged, divided by all within-sequence adjacent-frame opportunities. Transition delay is the number of classification frames from a ground-truth transition to the first matching prediction, restricted to the target terrain segment. A transition is missed if no matching prediction occurs before the next terrain change or sequence end; miss rate is the fraction of all transitions missed. Because mean delay includes only matched transitions, it must be interpreted jointly with miss rate. These metrics do not characterize anticipation or sustained switching, which are examined separately in the closed-loop analysis.

Table II and Fig. 4(a) show that temporal processing improves aggregate recognition and suppresses spurious switching for both representations. Relative to instantaneous selection, Bayes increases balanced accuracy by 1.08 percentage points for geometric features and 1.53 points for raw depth, while reducing false-transition rates by approximately 58% and 61%, respectively. These improvements accompany longer matched delays and higher miss rates, with EMA providing an intermediate operating point. For geometric features, the additional gain over EMA is modest: 0.13 points in balanced accuracy and a 0.49-point reduction in false-transition rate, at the cost of 0.23 additional delay frames and a 0.23-point increase in miss rate. Across matched selectors, geometric features achieve higher mean balanced accuracy

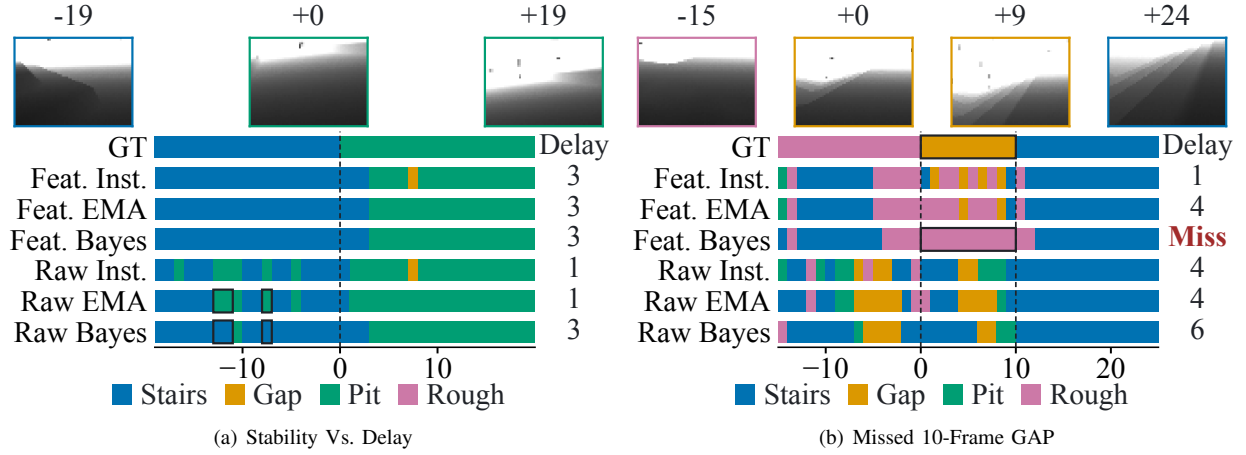


Fig. 5. Sequential prediction stability and transition responsiveness.

and lower mean false-transition rates, delays, and miss rates.

The response distributions in Fig. 4(b) expose differences obscured by mean matched delay. Instantaneous raw-depth predictions match more transitions immediately than geometric features (31.0% versus 22.6%), but fewer within two frames (79.5% versus 87.7%). Under Bayes, the corresponding within-two-frame fractions are 61.4% and 77.6%. Thus, greater agreement at the boundary can coexist with more delayed and missed responses. Including all transitions in these distributions avoids conditioning the comparison on successful matches; immediate agreement alone does not establish anticipatory or sustained switching.

The accuracy gains from stronger persistence are concentrated in steady-state observations. Compared with EMA, Bayes increases feature-based steady-state accuracy from 98.40% to 98.60%, while reducing transition-window accuracy from 80.72% to 79.95%. Raw depth follows the same pattern: steady-state accuracy increases from 97.35% to 97.76%, while transition-window accuracy decreases from 74.04% to 73.48%. Figure 5(a) illustrates this tradeoff on an individual sequence. The outlined intervals identify three steady-state frames where raw-depth EMA incorrectly selects Pit on Stairs, whereas Bayes retains Stairs. At the subsequent transition to Pit, however, Bayes first matches after three frames, compared with one for EMA.

Missed transitions are concentrated in short target segments. Segments containing at most ten classification frames constitute 7.26% of transitions but account for 94.8% of feature-based Bayes misses and 90.8% of raw-depth Bayes misses. Figure 5(b) illustrates a complete ten-frame Gap: feature-based Bayes retains Rough throughout, whereas instantaneous selection and EMA first match after one and four frames. Raw-depth Bayes matches after six frames, showing that the same short segment is not missed by every filtered classifier. These selected examples illustrate how temporal persistence interacts with perceptual evidence; they do not establish the prevalence or control consequences of these behaviors. Despite its lower aggregate offline accuracy, raw depth therefore enables recognition of a brief terrain change that feature-based Bayes misses. Such differences may matter disproportionately

near locomotion-critical obstacles, motivating the closed-loop investigation of whether transition-specific recognition and selection timing explain the discrepancy between offline and locomotion rankings. Together, the results demonstrate that improved aggregate recognition and switching stability can accompany delayed or missed terrain changes. Responsiveness therefore requires considering matched-delay distributions and misses jointly. The subsequent closed-loop experiment examines whether these offline tradeoffs predict specialist capability and the effectiveness of routing relative to distillation.

C. Closed-Loop Multi-Terrain Locomotion Performance

We compare routed specialists and the distilled student on matched mixed-terrain courses spanning easy, medium, and hard difficulty. Ground-truth routing and all six perception-selector combinations establish how routing design affects this comparison. We jointly assess locomotion capability and the additional resources required after specialist acquisition.

Locomotion Performance We compare six routers, ground-truth routing, and distillation from the same specialist library on matched rough, gap, stair, and climbing courses, using 900 episodes per method across three difficulties and three evaluation seeds. Metrics include episodic success (completion or timeout survival), forward distance, and terrain-label disagreement. Five of six routers outperform distillation’s 74.3% success rate (Table III). Raw Depth + Bayes leads all routed configurations with 92.6% success and 19.48 m distance, compared with 99.1% and 20.62 m for ground-truth routing. On hard terrain, it achieves 84.7% success, versus 79.7% for distillation and 69.3% for the strongest feature-based routers.

Temporal processing improves both representations, with EMA strongest for features and Bayes strongest for raw depth. Despite superior offline recognition, feature-based routers consistently underperform raw-depth routers in locomotion. Features + Bayes also achieves the lowest disagreement (42.4%), yet substantially lower success than Raw Depth + Bayes (43.1% disagreement). Skill-switch rates decrease from 9.39% with instantaneous selection to 4.31% with EMA and 3.64% with Bayes for features, and from 8.96% to 4.82% and 3.85%, respectively, for raw depth; while ground-truth routing

TABLE III

MULTI-TERRAIN CLOSED-LOOP PERFORMANCE OVER 900 EPISODES PER METHOD. EPISODIC SUCCESS INCLUDES COURSE COMPLETION OR SURVIVAL UNTIL TIMEOUT. SKILL-SWITCH RATE IS THE NUMBER OF SWITCHES PER 100 CLASSIFICATION UPDATES, AVERAGED ACROSS RUNS. BEST ROUTED VALUES ARE BOLD.

Method	Episodic success (%) $\uparrow$				Aggregate outcomes		
	Easy	Medium	Hard	Aggregate	Forward distance (m) $\uparrow$	Wrong skill (%) $\downarrow$	Skill-switch rate (%) $\downarrow$
Ground-truth routing	99.0	99.3	99.0	99.1	20.62	0.0	1.24
Features + Instantaneous	74.7	77.7	61.3	71.2	16.93	43.8	9.39
Features + EMA	83.0	86.7	69.3	79.7	17.96	43.8	4.31
Features + Bayes	83.7	85.0	69.3	79.3	18.01	42.4	3.64
Raw depth + Instantaneous	90.0	89.7	82.7	87.4	18.71	44.0	8.96
Raw depth + EMA	95.7	91.3	83.3	90.1	19.27	43.6	4.82
Raw depth + Bayes	98.0	95.0	84.7	92.6	19.48	43.1	3.85
Distilled policy	75.3	78.0	69.7	74.3	17.21	— [†]	— [†]

[†]Undefined: the distilled policy does not emit an explicit skill label.

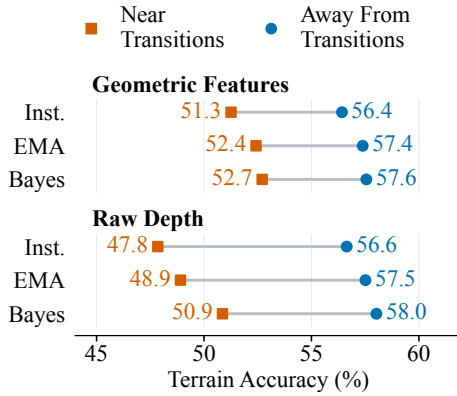


Fig. 6. Classification accuracy near and away from terrain transitions on shared closed-loop observations.

TABLE IV
BALANCED ACCURACY (%) FOR OFFLINE REFERENCES AND
SAME-TRAJECTORY REPLAY.

Representation	Selector	Offline	Rollout source		
			Oracle	F+B	R+B
Features	Inst.	94.75	86.68	60.25	62.25
	EMA	96.04	89.86	61.41	63.70
	Bayes	96.28	90.73	61.60	63.87
Raw depth	Inst.	92.62	81.70	60.61	62.16
	EMA	94.08	84.07	62.01	63.40
	Bayes	94.63	85.14	62.66	64.02

switches on 1.24% of classification updates along its own trajectories. These results support integration without further locomotion-policy optimization while showing that terrain-label agreement alone does not predict control performance. We therefore use same-trajectory replay to investigate the recognition-ranking reversal and the coexistence of high disagreement with reliable locomotion.

From Offline Recognition to Closed-Loop Routing Same-trajectory replay tests whether offline recognition rankings persist on observations encountered during control. Table IV shows that the feature-based Bayesian classifier’s advantage on oracle trajectories (90.73% versus 85.14% balanced

TABLE V
POST-SPECIALIST INTEGRATION AND DEPLOYMENT COSTS.

Metric	Features	Raw Depth	Distilled
Total simulator transitions	19,000,000	19,000,000	720,000,000
Collection / teacher labeling (min)	75.74	75.74	510.41 $\pm$ 3.79
Offline preparation (min)	50.85	2.08	0.00
Fitting / optimization (min)	8.62 $\pm$ 4.49	38.00 $\pm$ 4.39	26.93 $\pm$ 5.21
Total integration time (min)	135.21 $\pm$ 4.57	115.82 $\pm$ 3.97	537.34 $\pm$ 8.82
Allocated GPU-hours	1.41 $\pm$ 0.09	1.90 $\pm$ 0.10	8.97 $\pm$ 0.04
Trainable parameters	192,900	1,149,540	1,154,737
Deployable checkpoint size (MB)	37.99	41.71	9.33

accuracy) disappears on Features + Bayes trajectories (61.60% versus 62.66%). On shared observations pooled across routed trajectories, Fig. 6 reports 56.4–58.0% accuracy outside transition windows and 47.8–52.7% within them. Temporal filtering improves both regions, with geometric features retaining higher accuracy near transitions and raw depth slightly higher accuracy elsewhere. These results are consistent with a shift from the oracle-routed training distribution: different skill selections and switching times may alter body motion and camera viewpoints, exposing classifiers to underrepresented depth and robot-state observations.

Terrain-label disagreement alone does not predict locomotion failure: successful Raw Depth + Bayes episodes still contain 43.3% disagreement, suggesting that some alternative skill selections are tolerable. Features + EMA also increases success from 71.2% to 79.7% while disagreement remains approximately 43.8%, accompanied by a 54% reduction in skill-switch rate. These findings suggest that when errors occur, which skills are selected, and how frequently selections change matter alongside classification accuracy.

D. Integration and Deployment Costs

Table V reports integration costs on an RTX 4090, excluding original specialist training and evaluation. Distillation requires approximately 38 $\times$ more simulator transitions and 4.0–4.6 $\times$ the integration time of routing, with rollout collection and teacher labeling accounting for 95% of its duration. Although the feature MLP fits faster, geometric preprocessing increases its total integration time to 135.21 min, compared with 115.82 min for raw depth. Routing also requires fewer allocated GPU-hours, but retains a larger deployment package: 37.99–41.71 MB including the specialist library, versus

TABLE VI
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Representation	Selector	Total Routing (ms)	Classifier (ms)	Temporal Selection (ms)	RSS (MiB)
Features	Instantaneous	5.72 ± 0.54	5.44 ± 0.56	0.12 ± 0.23	321.35 ± 0.52
	EMA	6.19 ± 0.56	5.72 ± 0.55	0.31 ± 0.35	320.53 ± 1.01
	Bayes	6.30 ± 0.56	5.55 ± 0.53	0.59 ± 0.44	323.26 ± 1.03
Raw depth	Instantaneous	7.67 ± 3.83	7.44 ± 3.80	0.08 ± 0.18	328.74 ± 0.13
	EMA	7.87 ± 4.44	7.41 ± 4.25	0.27 ± 0.74	329.15 ± 0.15
	Bayes	7.83 ± 4.34	7.15 ± 3.96	0.53 ± 1.05	330.72 ± 1.14

9.33 MB for distillation.

Table VI reports CPU inference on a Jetson Orin NX, using 50 warmup calls and 1,000 measurements per pipeline (mean ± sample standard deviation). Feature-based routing is faster and less variable than raw depth (5.72–6.30 versus 7.67–7.87 ms), with classifier processing dominating runtime and selection contributing less than 0.6 ms on average. Process RSS is similar across pipelines (approximately 320–331 MiB). These timings exclude camera processing and specialist-policy execution. The comparison therefore favors raw depth for integration time and geometric features for recurring router cost. distillation provides a smaller deployment package and additional potential deployment savings through not having any router overhead; come at a substantial training cost.

V. CONCLUSION

We investigated lightweight perceptual routing as an alternative to distilling an existing library of frozen locomotion specialists. In simulation, raw-depth routing with Bayesian smoothing outperforms the evaluated distilled controller while requiring substantially less integration effort and no additional locomotion-policy optimization. Learned depth representations yield stronger closed-loop performance than engineered features, while temporal smoothing suppresses spurious switching. Surprisingly, stronger offline classification does not predict better locomotion, and successful routed rollouts contain substantial terrain-label disagreement outside transition windows, highlighting the unequal control consequences of perception errors. These findings support simple routing as a practical integration strategy, but remain an exploratory simulation result: onboard inference measurements characterize computational costs without establishing hardware locomotion performance. Future work will improve the sim-to-real transfer of the specialist policies and evaluate whether these routing benefits and error-tolerance patterns persist during physical robot locomotion.

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